

ASSIGNMENT RNA-Seq Guided Replication Using Galaxy and GitHub

Group Topic

Group 3 - Infection

Group Members

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Part 3. Submit the Study for Approval

Item	Group Information
Full title of the paper	RNA-Seq Analysis of the Host Response to <i>Staphylococcus aureus</i> Skin and Soft Tissue Infection in a Mouse Model.
Authors	Brady RA, Bruno VM, Burns DL
Year	2015
Journal	PLOS ONE
Organism	Mouse (BALB/c, 10-week-old female)
Tissue or cell type	Ear skin/tissue
Assigned stress topic	Infection
Controlcondition	Uninfected ear from the same challenged mouse; for the systemic response, ears from naive mice were used as the control
Treatment/stress condition	Epicutaneous infection with <i>Staphylococcus aureus</i> USA300 strain SAP149.
Numberofbiologicalreplicates	3 naive and 3 infected mice per time point (Days 1,4, and 7)
RNA-seq repository	NCBI Sequence Read Archive (SRA)
BioProject/Studyaccession	PRJNA239895
Runaccessionnumbersthat the group plans to use	41 SRR1193981 - Control 33 SRR1193913 - Control 37 SRR1193959 - Infected 29 SRR1193865 - Infected

Referencegenomeaccession	GCF_000001635.27
Annotationsource	NCBI Ref-seq
Approximate size of the selected	376.01Gb
Short description of the authors' RNA-seq analysis pipeline	RNA was poly(A)-enriched and sequenced as 100-bp paired-end reads using the Illumina HiSeq platform. Reads were aligned to the UCSC BALB/c mouse reference genome using TopHat, followed by gene-level read counting and differential expression analysis using DESeq. Genes with FDR <0.01 and at least 2-fold change were considered differentially expressed.

Part 5&6. Import the RNA-Seq Reads & Import the RNA-Seq Reads

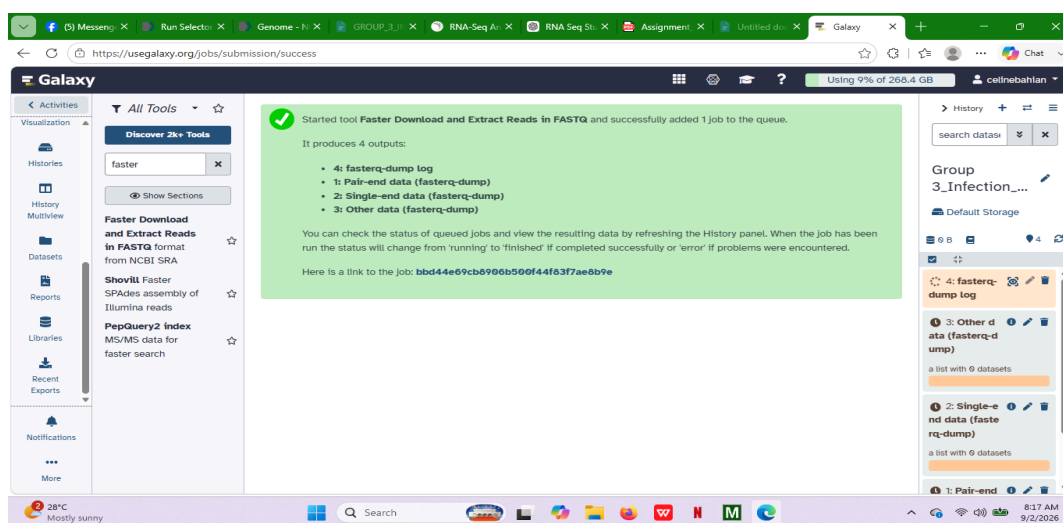


Figure 1. Downloading paired-end RNA-seq FASTQ reads from the NCBI Sequence Read Archive (SRA) into Galaxy using the Download and Extract Reads in FASTQ tool. Two control samples (SRR1193981 and SRR1193913) and two Staphylococcus aureus–infected samples (SRR1193959 and SRR1193865) were imported, generating forward (R1) and reverse (R2) FASTQ files for downstream quality assessment and transcriptomic analysis.

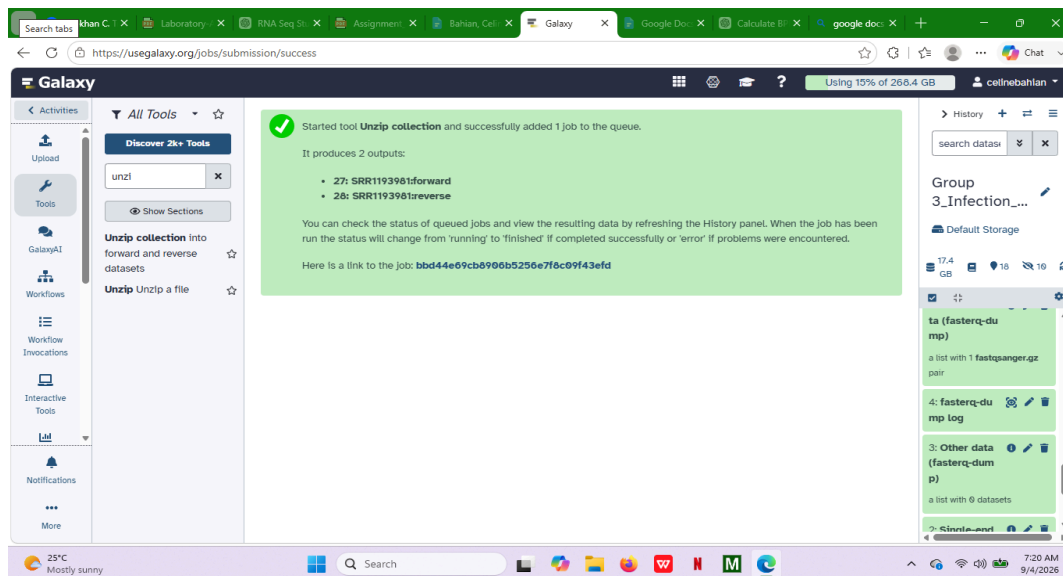


Figure 2. Renaming and updating dataset attributes in the Galaxy history after importing the RNA-seq reads. The paired-end FASTQ files were assigned descriptive identifiers (Control_R1_rep1, Control_R2_rep1, Control_R1_rep2, Control_R2_rep2, Infected_R1_rep1, Infected_R2_rep1, Infected_R1_rep2, and Infected_R2_rep2) to clearly distinguish control and *Staphylococcus aureus*-infected biological replicates for downstream RNA-seq analysis.

Part 7. RNA-Seq Quality Control

	Number of reads	General sequence quality	Adapter contamination	Over represented sequences	Any obvious quality problem
Control_R1_rep1	24,216,915	HIGH (PASS)	NONE DETECTED	NONE DETECTED	NO
Control_R2_rep1	24,216,915	HIGH (PASS)	NONE DETECTED	NONE DETECTED	NO
Control_R1_rep2	24,216,915	HIGH (PASS)	NONE DETECTED	NONE DETECTED	NO
Control_R2_rep2	25,787,474	HIGH (PASS)	NONE DETECTED	NONE DETECTED	NO
Infected_R1_rep1	23,817,739	HIGH (PASS)	NONE DETECTED	NONE DETECTED	NO
Infected_R2_rep1	23,817,739	HIGH (PASS)	NONE DETECTED	NONE DETECTED	NO

Infected_R1_rep2	23,817,739	HIGH (PASS)	NONE DETECTED	NONE DETECTED	NO
Infected_R2_rep2	21,517,900	HIGH (PASS)	NONE DETECTED	NONE DETECTED	NO

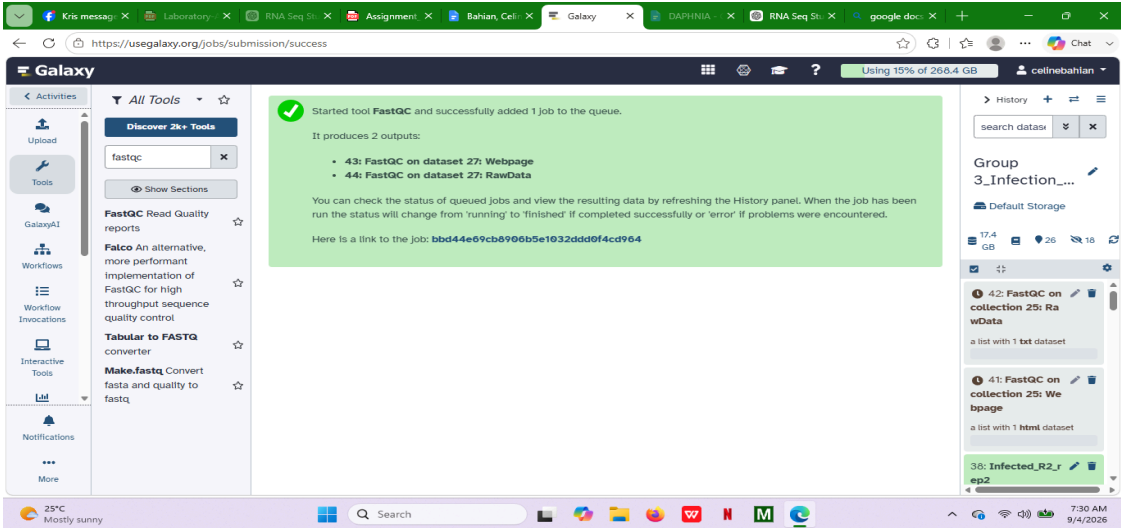


Figure 3. FastQC job successfully submitted in Galaxy for the selected paired-end RNA-seq sample, generating both the Webpage and RawData outputs for quality assessment.

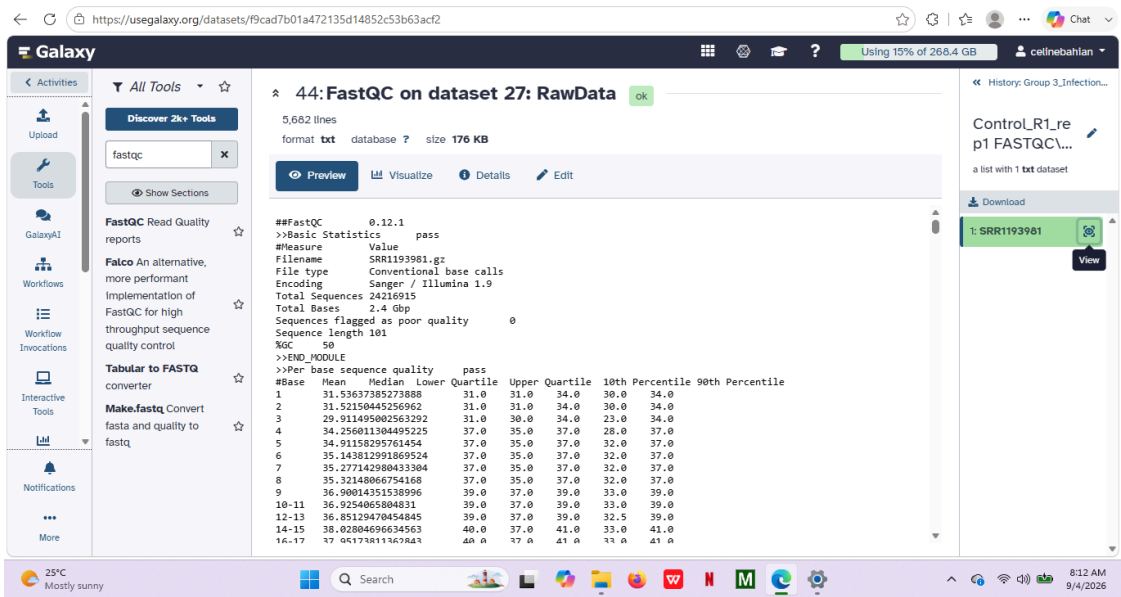


Figure 4. FastQC raw-data results for Control_R1_Rep1, showing the quality statistics and per-base sequence quality of the raw reads before quality filtering.

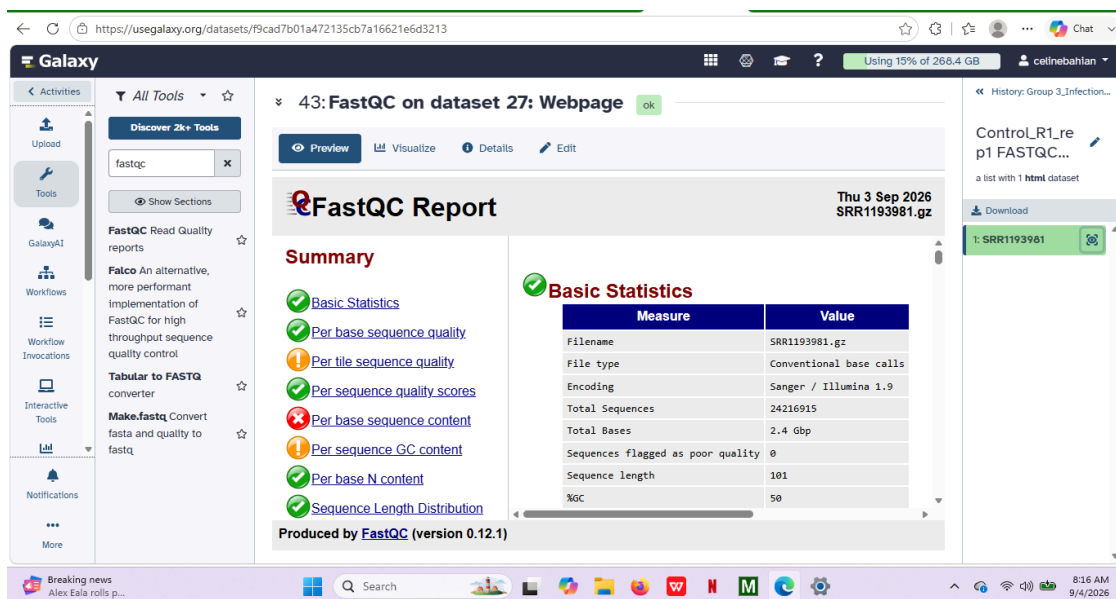


Figure 5. FastQC report webpage for Control_R1_Rep1 showing the quality assessment of the raw sequencing reads, including basic statistics, per-base sequence quality, sequence quality scores, and other quality-control metrics.

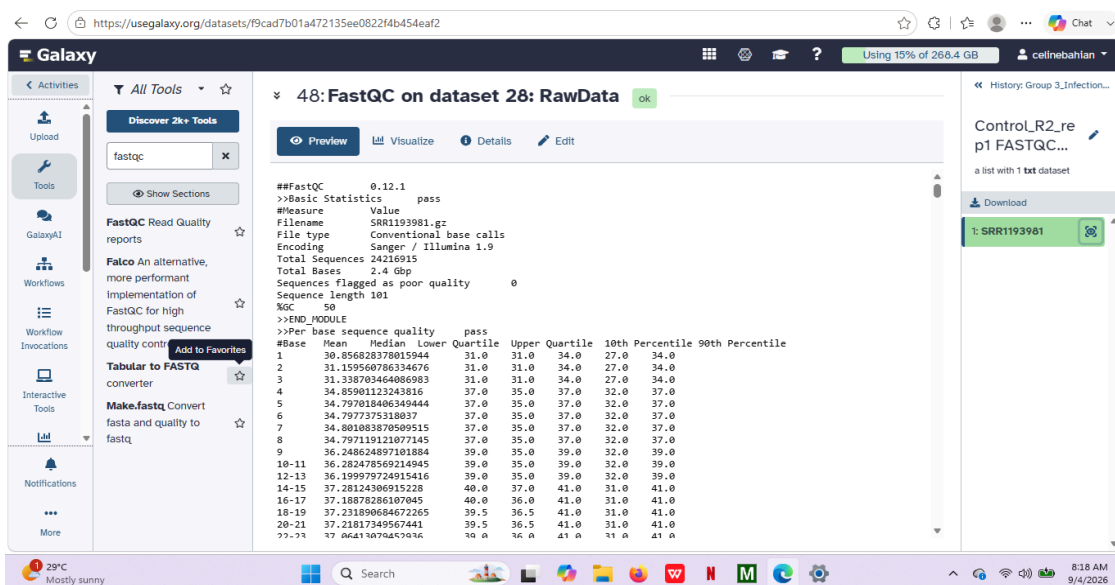


Figure 6. FastQC raw-data results for Control_R1_Rep2, showing the quality statistics and per-base sequence quality of the raw reads before quality filtering.

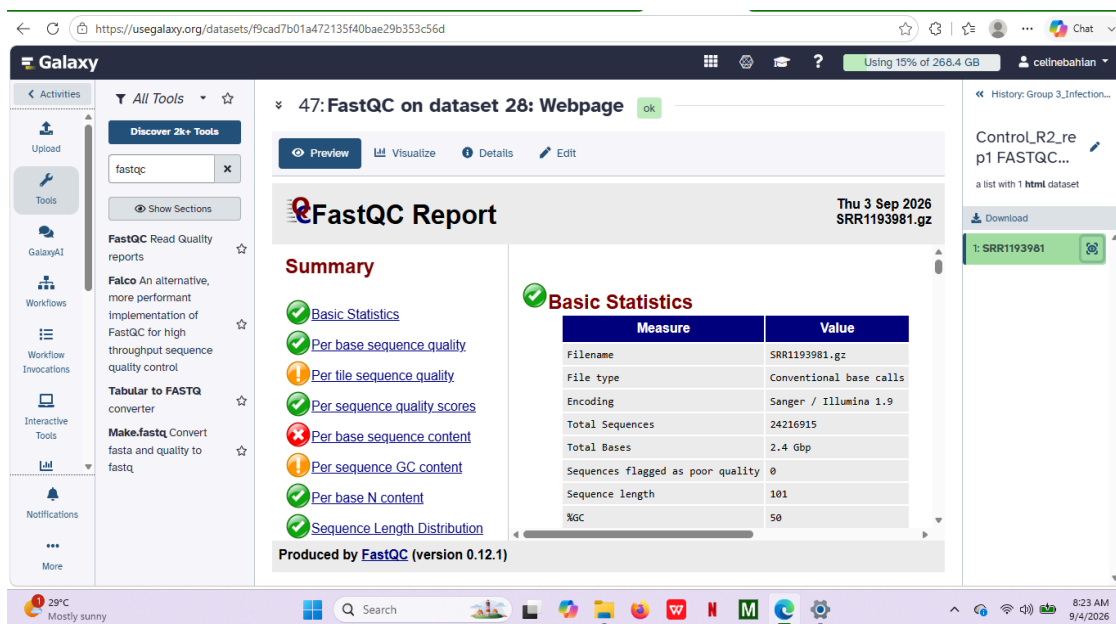


Figure 7. FastQC report webpage for Control_R1_Rep2 showing the quality assessment of the raw sequencing reads, including basic statistics, per-base sequence quality, sequence quality scores, and other quality-control metrics.

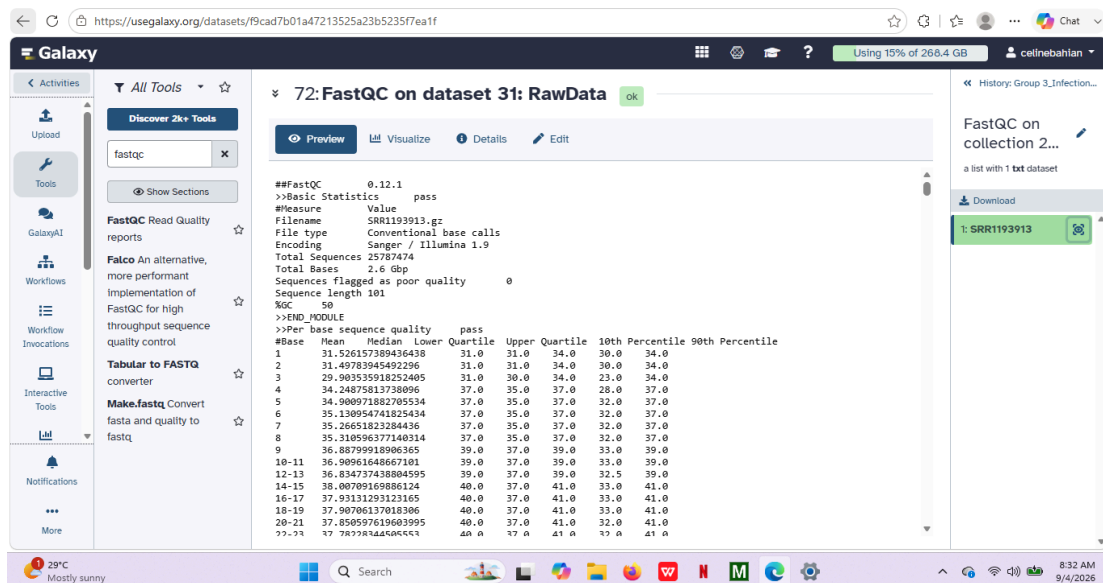


Figure 8. FastQC raw-data results for Control_R2_Rep1, showing the quality statistics and per-base sequence quality of the raw reads before quality filtering.

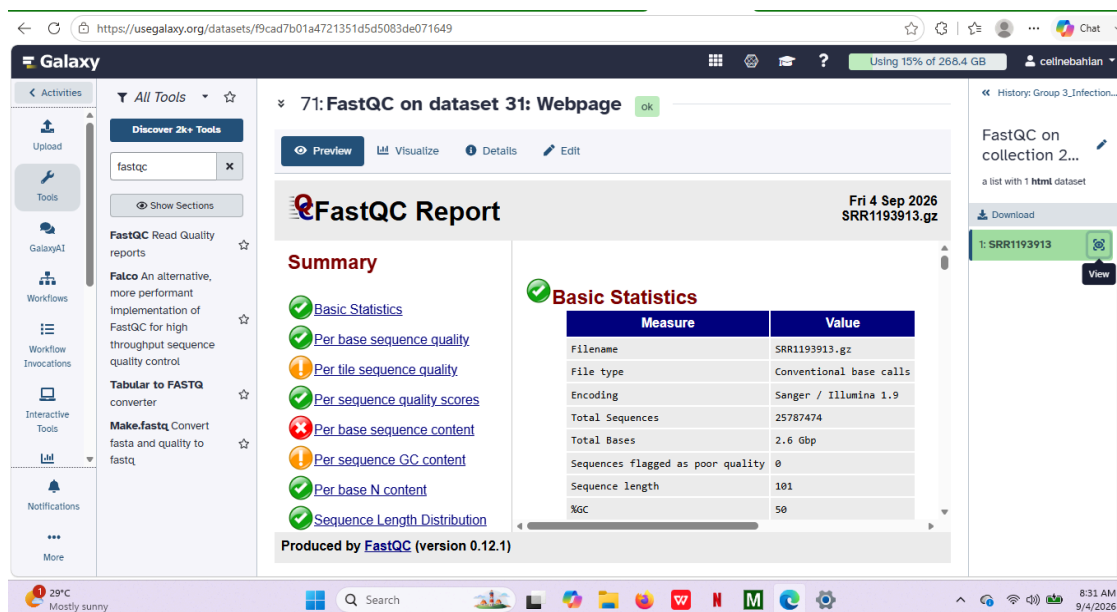


Figure 9. FastQC report webpage for Control_R2_Rep1 showing the quality assessment of the raw sequencing reads, including basic statistics, per-base sequence quality, sequence quality scores, and other quality-control metrics.

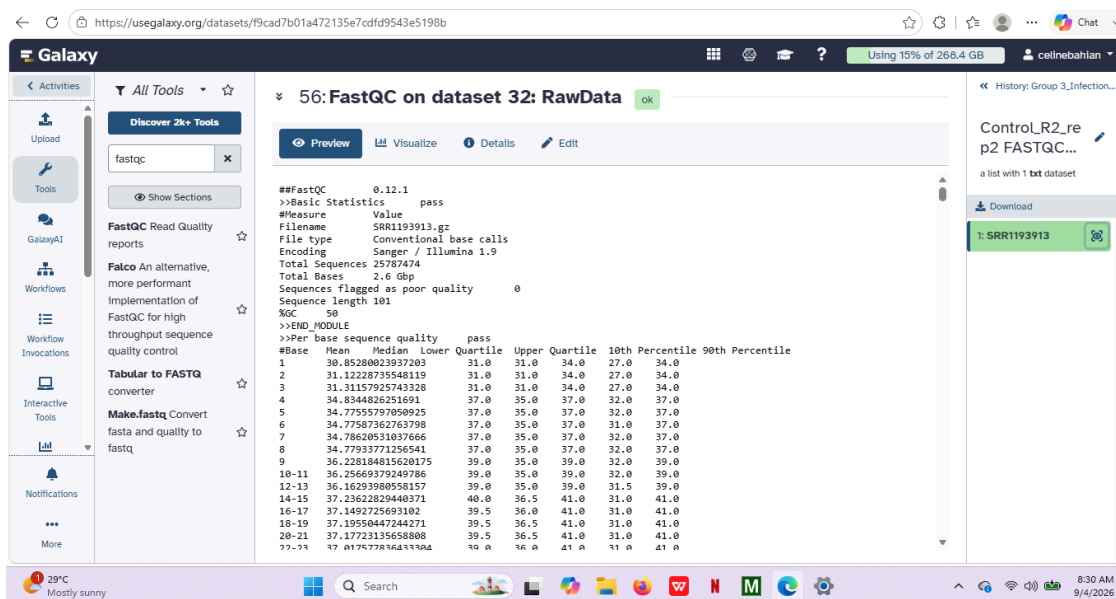


Figure 10. FastQC raw-data results for Control_R2_Rep2, showing the quality statistics and per-base sequence quality of the raw reads before quality filtering.

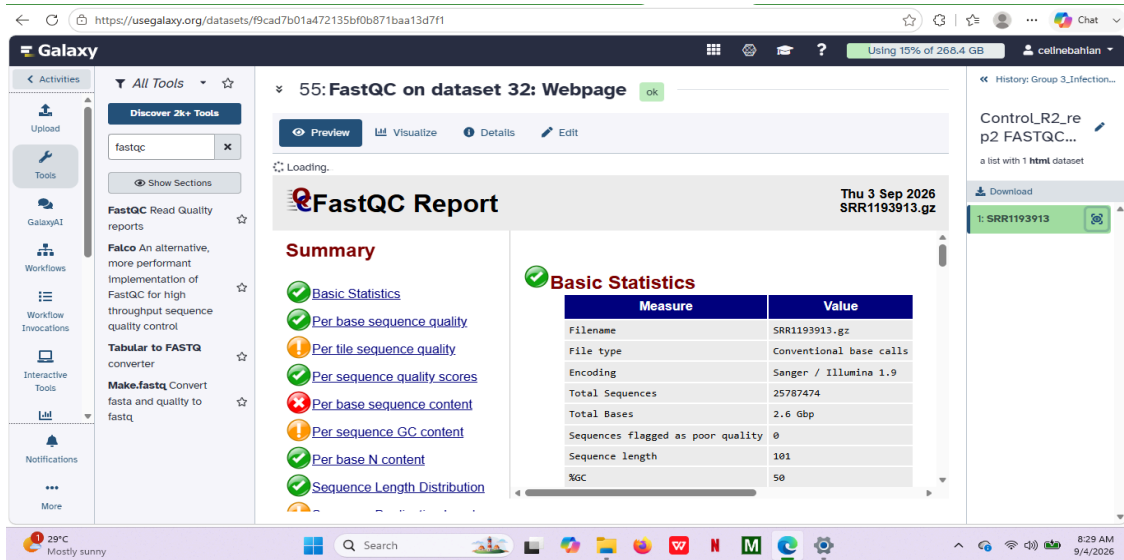


Figure 11. FastQC report webpage for Control_R2_Rep2 showing the quality assessment of the raw sequencing reads, including basic statistics, per-base sequence quality, sequence quality scores, and other quality-control metrics.

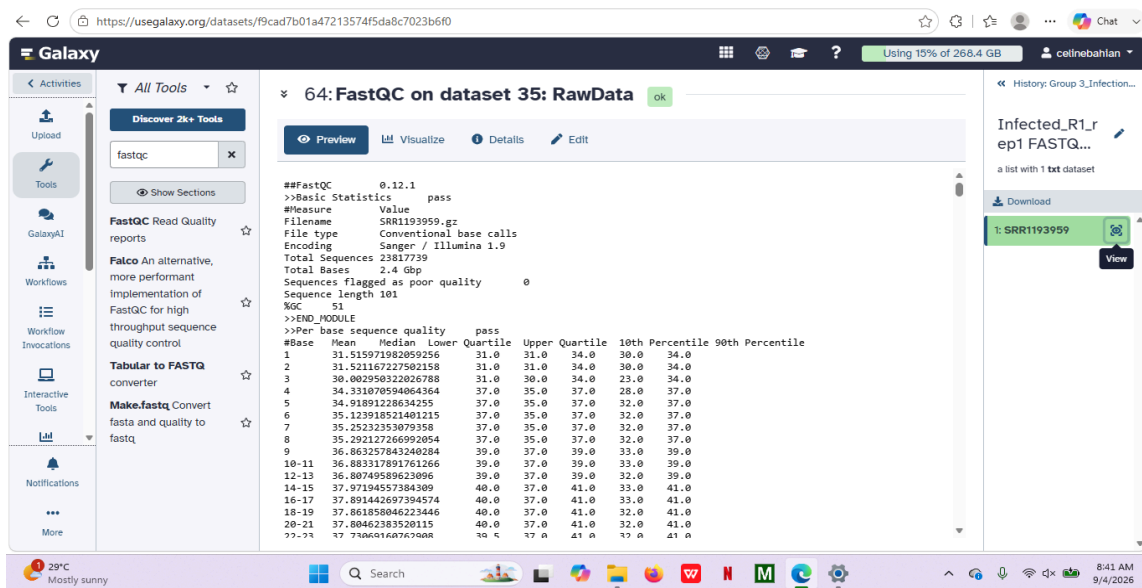


Figure 12. Raw FastQC data of Infected_R1_Rep1 showing the basic statistics and quality scores of the sequencing reads before quality filtering.

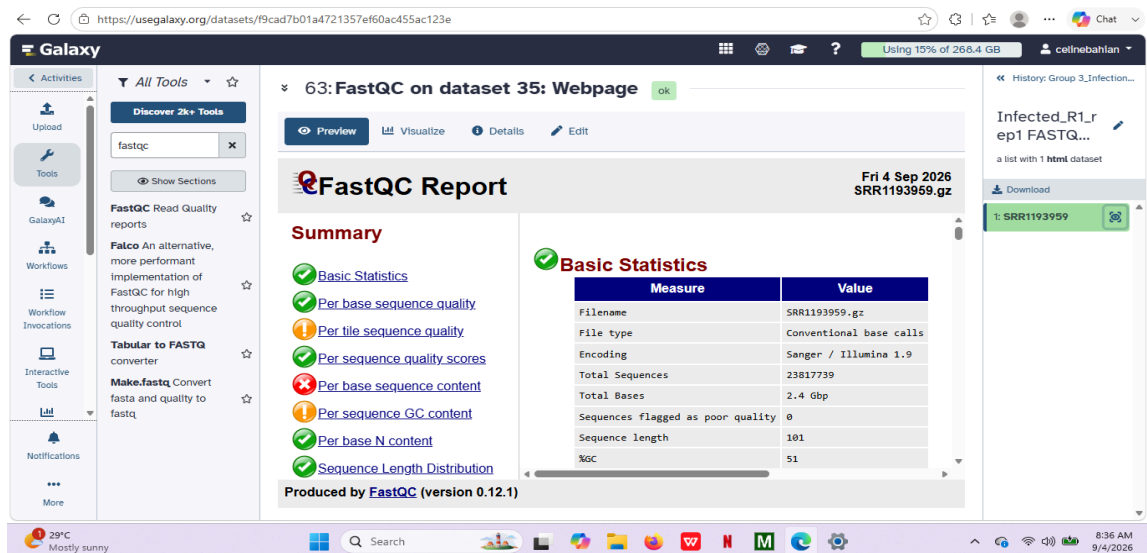


Figure 13. FastQC report webpage for Infected_R1_Rep1 showing the quality assessment of the raw sequencing reads, including basic statistics, per-base sequence quality, sequence quality scores, and other quality-control metrics.

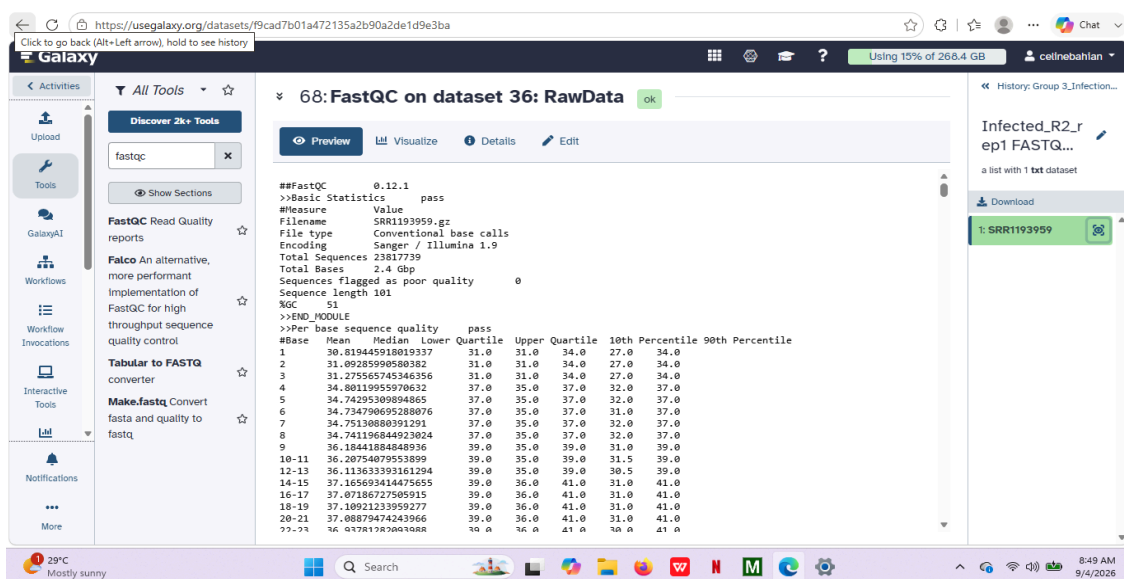


Figure 14. Raw FastQC data of Infected_R1_Rep2 showing the basic statistics and quality scores of the sequencing reads before quality filtering.

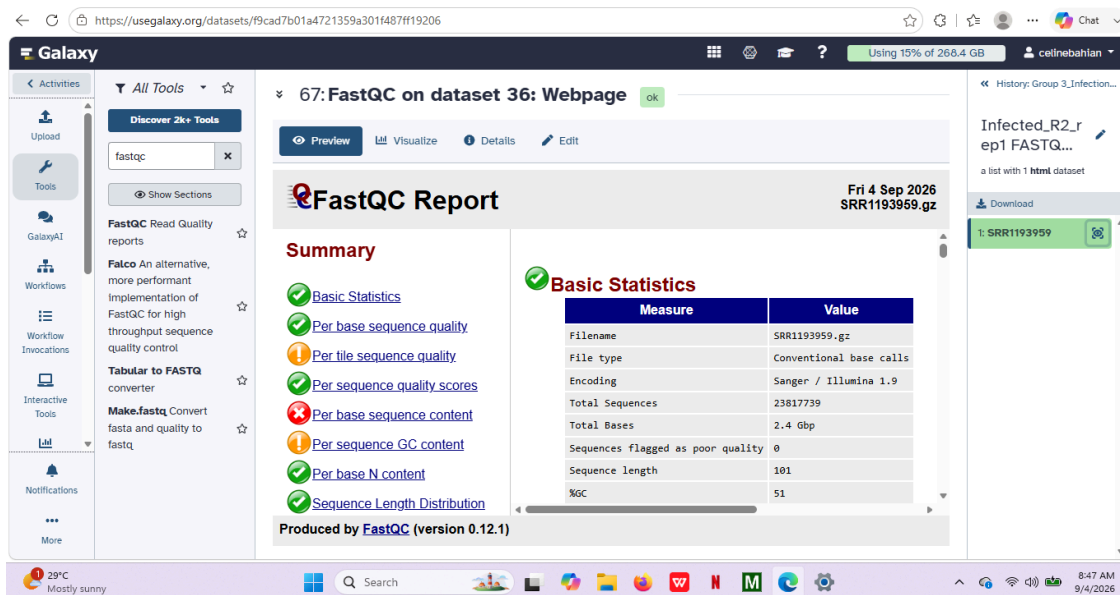


Figure 15. FastQC report webpage for Infected_R1_Rep2 showing the quality assessment of the raw sequencing reads, including basic statistics, per-base sequence quality, sequence quality scores, and other quality-control metrics.

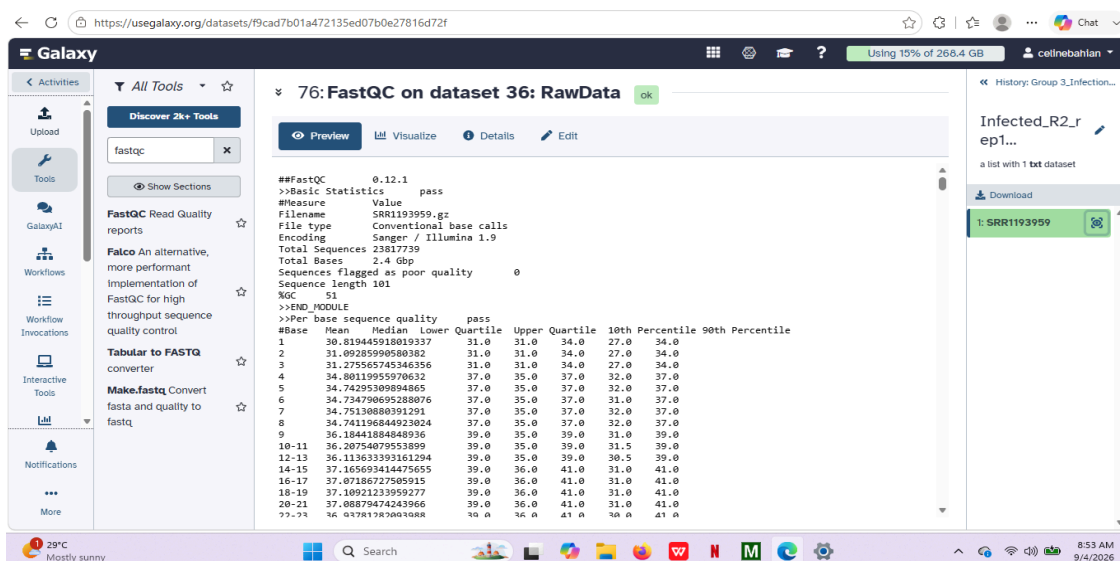


Figure 16. Raw FastQC data of Infected_R2_Rep1 showing the basic statistics and quality scores of the sequencing reads before quality filtering.

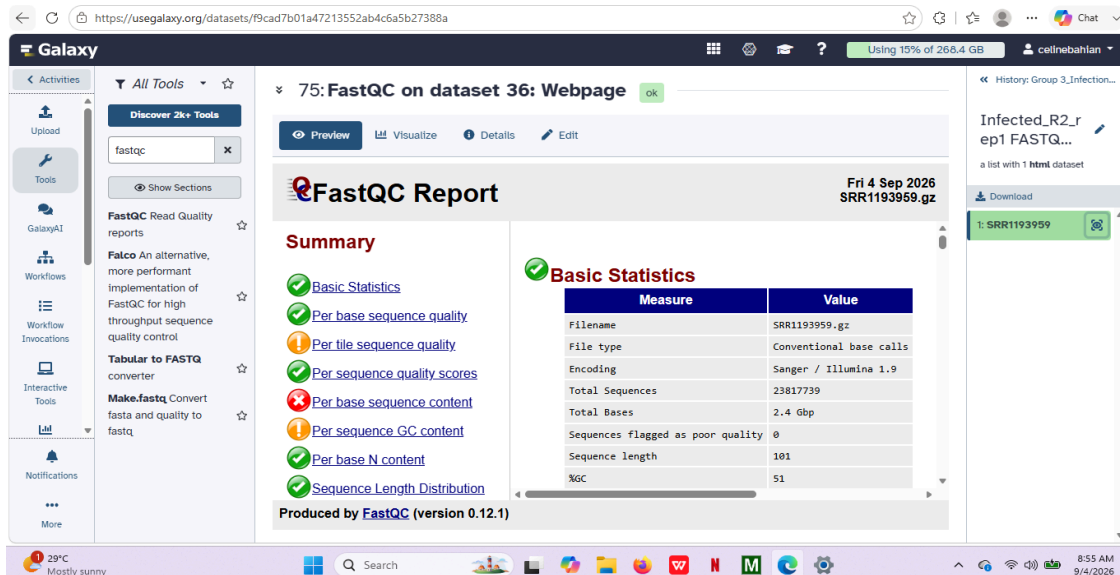


Figure 17. FastQC report webpage for Infected_R2_Rep1 showing the quality assessment of the raw sequencing reads, including basic statistics, per-base sequence quality, sequence quality scores, and other quality-control metrics.

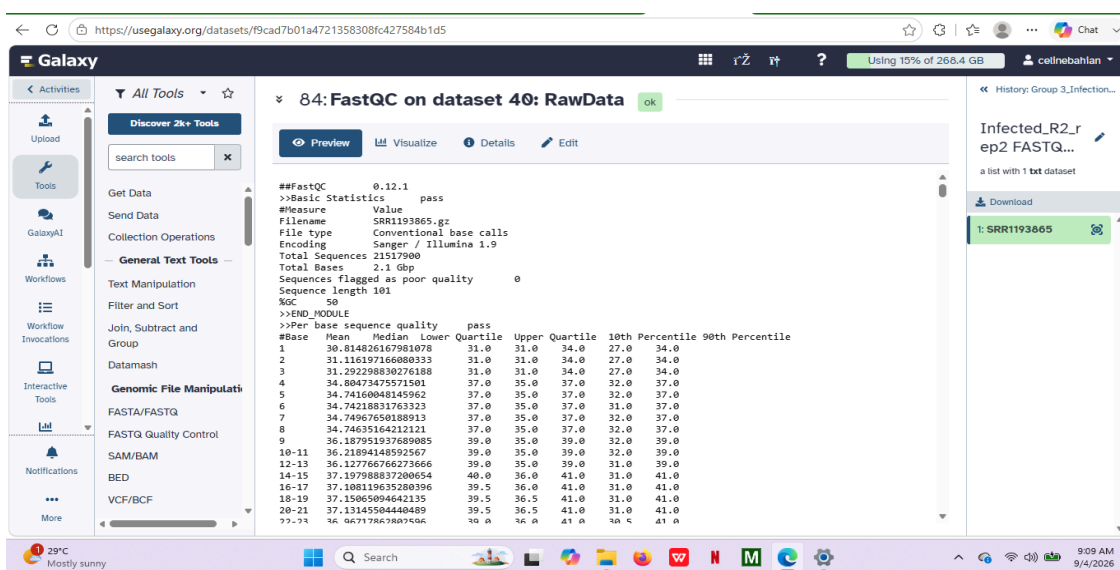


Figure 18. Raw FastQC data of Infected_R2_Rep2 showing the basic statistics and quality scores of the sequencing reads before quality filtering.

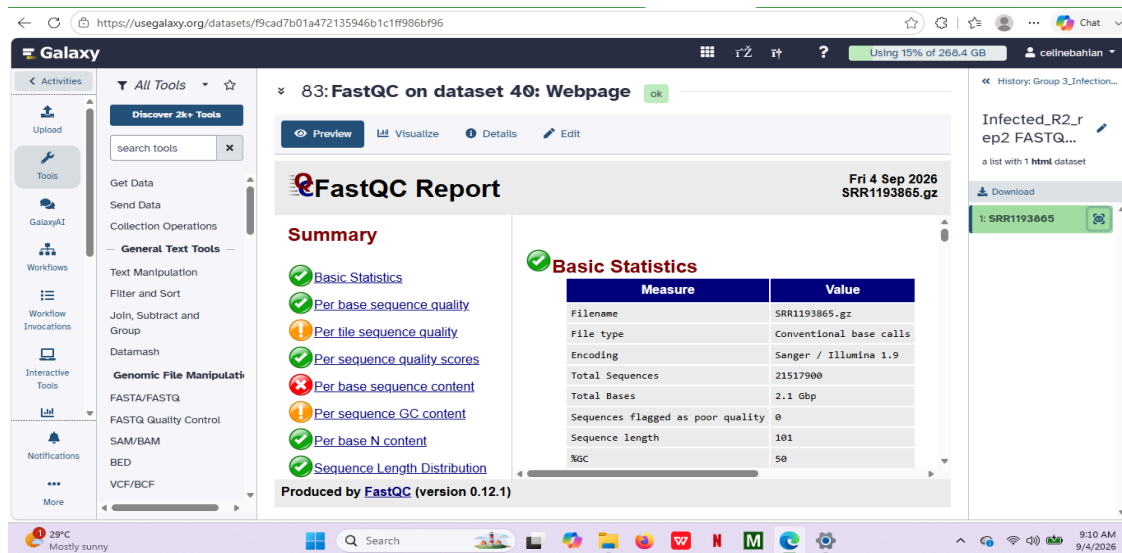


Figure 19. FastQC report webpage for Infected_R2_Rep2 showing the quality assessment of the raw sequencing reads, including basic statistics, per-base sequence quality, sequence quality scores, and other quality-control metrics.

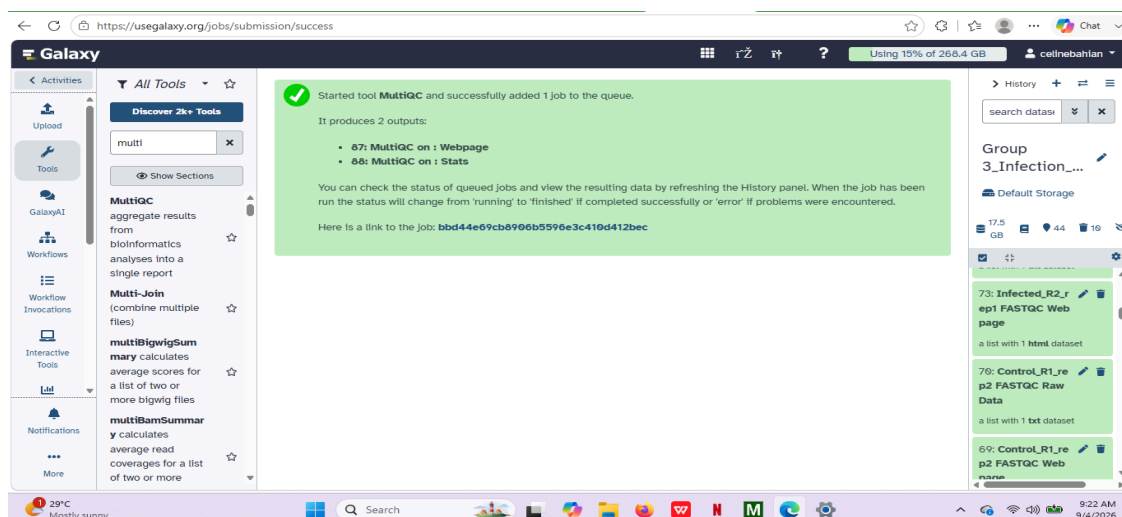


Figure 20. Successful execution of the MultiQC tool in Galaxy, showing the generated MultiQC outputs for the infected and control RNA-seq samples and confirming that the quality control reports were successfully processed and combined.

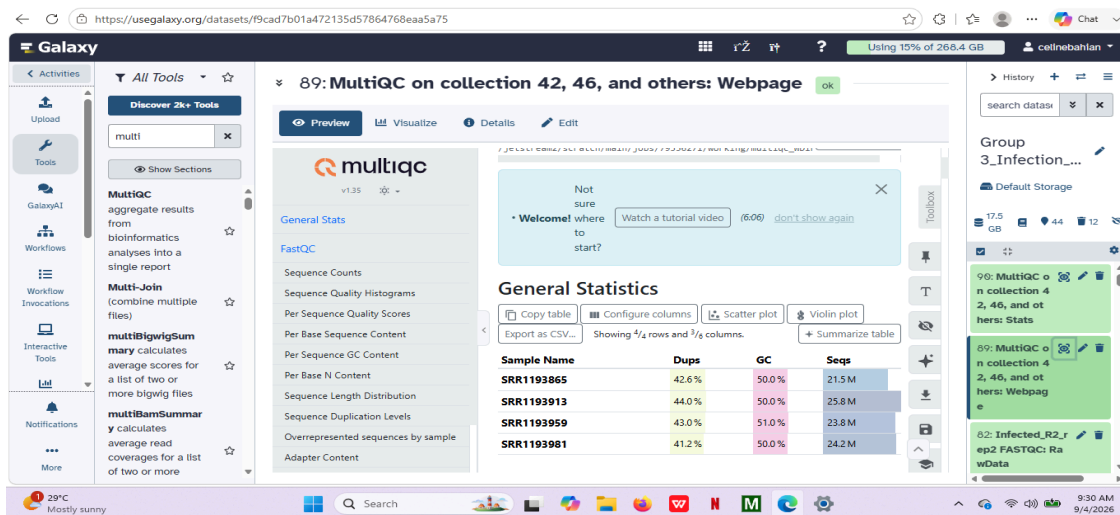


Figure 21. MultiQC report generated in Galaxy showing the general statistics of the RNA-seq samples, including GC content, duplication rate, and the total number of sequencing reads for each analyzed sample.

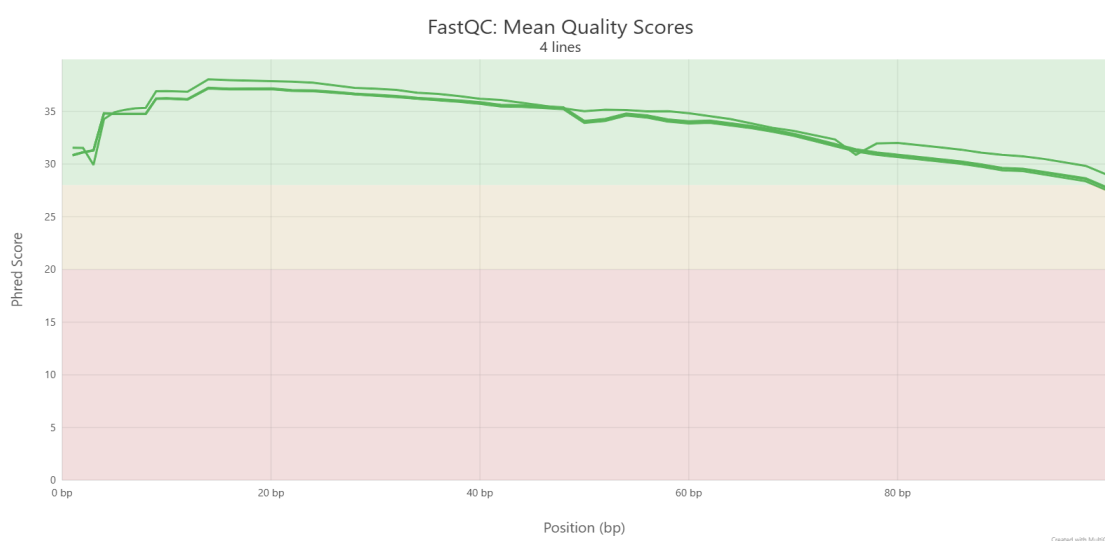


Figure 22. The MultiQC plot illustrates the mean quality scores of the RNA-seq reads across all base positions. Most quality scores remain above Q30, indicating that the sequencing reads are of consistently high quality.

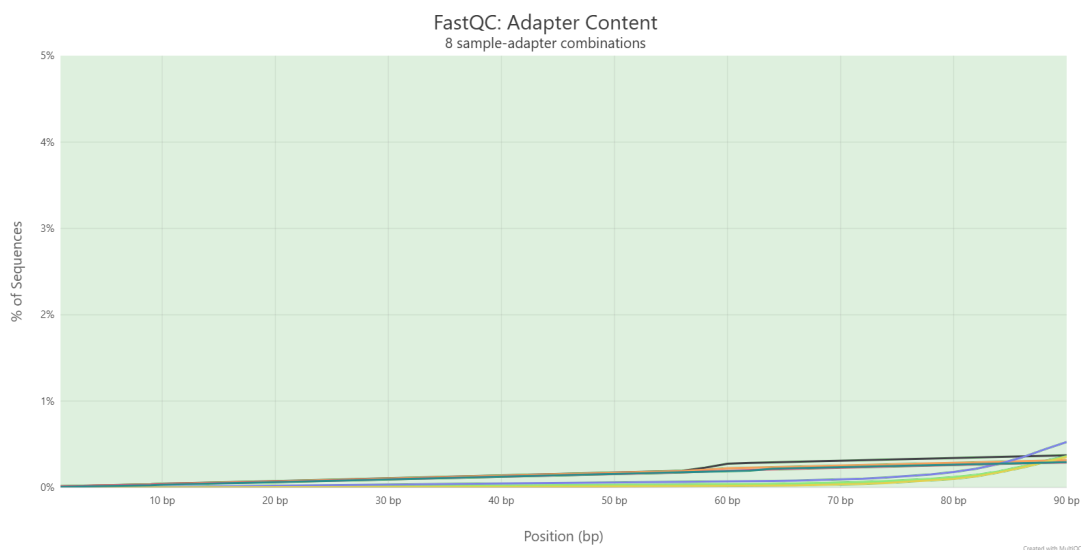


Figure 23. The MultiQC plot shows the adapter content across all RNA-seq samples at different base positions. Adapter contamination is consistently very low, indicating that the sequencing reads are generally clean and of good quality.

Part 9. Obtain the Reference Genome and Annotation

Item	Detail
Species	<i>Mus musculus</i>
Reference genotype	C57BL/6J
Assembly name	Mus_musculus_genome_v1
Assembly accession	GCA_000001635.9
Source	NCBI Genome FTP https://ftp.ncbi.nlm.nih.gov/genomes/all/GCF/000/001/635/GCF_000001635.27_GRCm39/
Genome FASTA file	Mus_musculus_v1_genome.fasta.gz (834.3 MB compressed; 61 sequences)
Annotation file	Mus-musculus_v1_annotation.gtf.gz (34.9 MB compressed)
FASTA file date	02/01/2024
GTF file date	02/01/2024
Downloaded	2026-09-04, via Galaxy Upload (Paste/Fetch from URL)
Galaxy dataset #	91 (FASTA), 92 (GTF)

Part 10. Map RNA-Seq Reads Using TopHat

Sample ID	Total reads	% Mapped	% Uniquely Mapped	Unusually low mapping
Control_rep1 (SRR1193981)	24,216,915	90.7%	NOT DETECTED	NO
Control_rep2 (SRR1193913)	25,787,474	90.4%	NOT DETECTED	NO
Stress24h_rep1 (SRR1193959)	23,817,739	89.6%	NOT DETECTED	NO
Stress24h_rep2 (SRR1193865)	21,517,900	89.9%	NOT DETECTED	NO

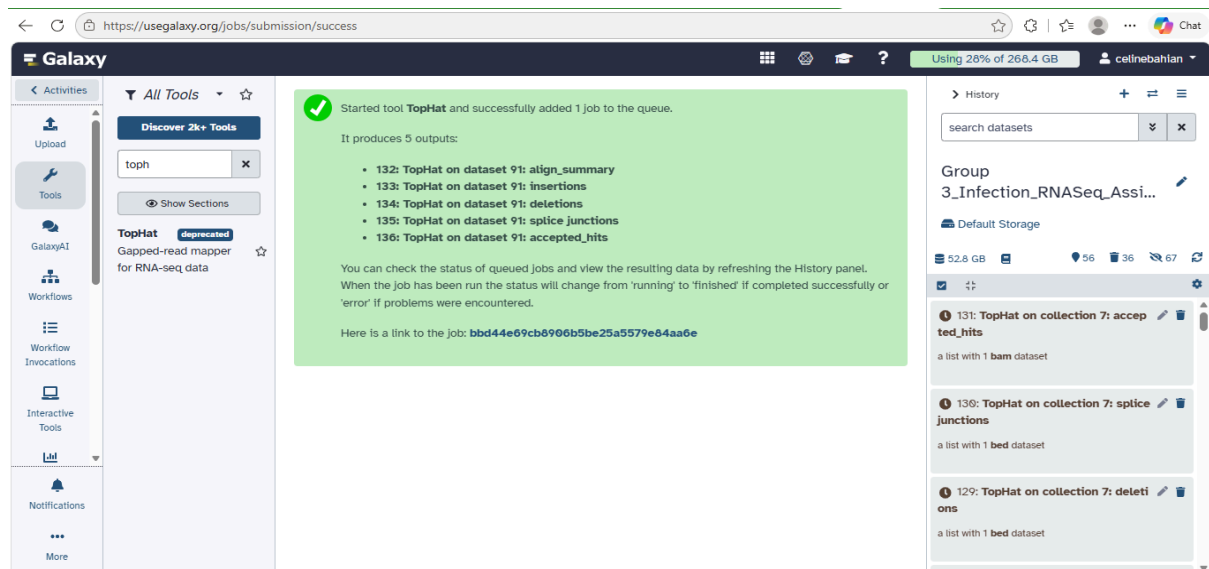


Figure 24. TopHat tool successfully started in Galaxy to map the RNA-seq reads to the reference genome and generate alignment outputs.

Part 11. Count Reads per Gene

Item	Detail
Counting tool	featureCounts (Galaxy)
Mapped RNA-seq files used	TopHat BAM outputs (4 samples: Control_rep1, Control_rep2, Infected_rep1, Infected_rep2), aligned to <i>Mus musculus</i> genome
Annotation file used	<i>Mus musculus</i> GTF (GCF_000001635.27 , matching genome used for alignment)
Feature type	exon (featureCounts default)

Gene identifier attribute	gene_id
Settings	Same settings applied to all 4 samples
Output	A complete table containing the 41,079 total gene counts is provided in a separate file.

Part 12. Differential Gene Expression

Positive log2 fold change	Gene expression increases under the treatment/stress condition.
Negative log2 fold change	Gene expression decreases under the treatment/stress condition.
Adjusted p-value < 0.05	This is generally considered evidence of a statistically significant difference in gene expression.
Adjusted p-value $\geq$ 0.05	The results do not provide sufficient statistical evidence for differential gene expression.

Output	Results
Total genes in output table	41,079
Number of genes tested	41,079
Number of significantly differentially expressed genes, if reported	3,792
Genes with positive log2 fold change.	1,422
Genes with negative log2 fold change.	2,370
Adjusted p-value or FDR threshold used.	<0.05

Output	Result
PCA plot	The PCA plot shows the separation of samples based on their gene-expression profiles. The Infected and Control samples form separate groups, with PC1 explaining 96% of the variance and PC2 explaining 3%.
Sample-to-sample distance	This plot shows the similarity or difference in gene-expression patterns between samples. Smaller distances indicate more similar samples, while larger distances indicate greater differences.

Dispersion estimates	The dispersion plot shows the estimated variability of gene counts relative to the mean of normalized counts. It includes individual gene estimates, the fitted trend, and final dispersion values.
p-value histogram	The histogram displays the distribution of p-values for the Control vs. Infected comparison. It shows the frequency of genes across p-value ranges from 0 to 1.
MA-plot	The MA-plot shows the relationship between the mean of normalized counts and log fold change, allowing genes with increased or decreased expression to be visualized.

Part 13. Select Genes for Biological Interpretation

Gene name / ID	log2FC	Adjusted p-value	Regulation	Known or Predicted Function	Possible Connection to Salinity Stress
Ccl4	-10.475	2.59×10^{-29}	Downregulated	Function needs annotation	Explain relationship to immune/inflammatory response
Acod1	-9.705	1.01×10^{-103}	Downregulated	Function needs annotation	Explain relationship to infection
Cxcl5	-9.491	6.08×10^{-47}	Downregulated	Function needs annotation	Explain relationship to infection
Cxcl2	-9.491	2.54×10^{-16}	Downregulated	Function needs annotation	Explain relationship to infection
Inmt	+3.979	3.27×10^{-10}	Upregulated	Function needs annotation	Explain relationship to infection

These were initially chosen because they have large changes in expression and very small adjusted p-values. However, because your assignment specifically mentions genes related to infection, we should also check their known functions before deciding that these are your Yes, some of the genes highlighted by the authors were also found in our analysis. This shows that our analysis was able to reproduce some of the original findings final five.

Part 14. Compare Your Results with the Original Study

ANSWERS:

1. Did your group recover the same general biological response described by the authors?
 - Yes, our analysis showed a similar overall biological response between the control and infected samples. The results showed clear changes in gene expression associated with infection.
2. Were any of the genes highlighted in the paper also identified in your analysis?
 - Yes, some of the genes highlighted by the authors were also found in our analysis. This shows that our analysis was able to reproduce some of the original findings.
3. Which results were similar?
 - The similar results included the identification of differentially expressed genes between the control and infected samples. Some genes also showed the same direction of expression as reported by the authors.
4. Which results were different?
 - Some genes identified by the authors were not among the significant genes in our analysis, while some of our identified genes were not highlighted in the paper. The magnitude of gene expression changes may also have differed.
5. Did you use exactly the same software as the authors?
 - No, our analysis was performed using tools available in Galaxy, such as featureCounts and DESeq2. The software and versions may not have been exactly the same as those used by the authors.
6. Did you use the same reference genome and annotation versions?
 - The available information does not confirm that our reference genome and annotation versions were exactly the same as those used by the authors. Differences in these files could affect gene counting and differential expression results.
7. Did you analyze all samples or only a subset?
 - We analyzed the samples provided for our Galaxy exercise rather than necessarily reproducing the complete sample set used by the authors. Using a different number of samples can affect the statistical results.
8. Could differences in software versions, parameters, sample number, or reference files explain differences in the results?
 - Yes, these differences could contribute to variations between our results and the authors' results. Changes in software, parameters, sample size, or reference files can affect read counting and differential expression analysis.
9. What did this exercise teach you about reproducibility in molecular biology?
 - This exercise showed us that reproducing RNA-seq results requires careful attention to the data, software, parameters, and reference files used. Even small differences in the analysis process can lead to different results.

Part 15. Galaxy Workflow Sharing

Galaxy workflow:

<https://usegalaxy.org/u/celinebahian/h/group-3-infection-rnaseq-assignment3>

Part 16. GitHub Group Repository

Github Link:

https://github.com/jilduhaylongsod/RNA_Seq_Guided_Replication_Group-3_Infection/setting

Part 17. Comparison Table

Analysis Step	Original Authors	Our Galaxy Re-Analysis
RNA-seq dataset	RNA-seq data from Staphylococcus aureus-infected mouse skin/tissue	4 samples: 2 Control and 2 Infected
Quality control	Not specified in the provided information	FastQC and MultiQC; reads showed generally high quality and low adapter contamination
Trimming	Not specified	Not indicated in the provided workflow
Reference genome	Mouse genome	GCF_000001635.27 / GRCm39
Annotation	NCBI RefSeq	NCBI RefSeq GTF
RNA-seq aligner	TopHat	TopHat
Gene counting	Not specified	featureCounts
Differential expression	Not specified	DESeq2
Significance threshold	Not specified	Adjusted p-value < 0.05
Main genes/pathways identified	Genes/pathways related to the host response to S. aureus infection	3,792 significantly differentially expressed genes; examples include Ccl4, Acod1, Cxcl5, Cxcl2, and Inmt

Part 18. Questions for Molecular Biology Understanding

ANSWERS:

1. Why can RNA-seq be used to study gene expression?
 - RNA-seq sequences the RNA molecules present in a sample. This allows us to determine which genes are being expressed and how much RNA is produced from each gene.
2. What is the difference between the genome FASTA and RNA-seq FASTQ files?
 - The genome FASTA contains the reference DNA sequence of the organism. The RNA-seq FASTQ files contain the actual RNA-derived sequencing reads from the samples, including their sequence and quality scores.
3. Why is a genome annotation needed for gene-level expression analysis?
 - The annotation tells us where genes and other features are located in the reference genome. It allows the RNA-seq reads to be assigned to specific genes so their expression can be counted.
4. What does a high number of reads for a gene generally suggest?
 - A high number of reads generally suggests that the gene is being expressed at a higher level in that sample. However, read counts need to be normalized before comparing expression between samples.
5. Why should biological replicates be included?
 - Biological replicates help show whether the observed gene-expression changes are consistent between different biological samples. They also improve the reliability of statistical analysis.
6. What is an upregulated gene?
 - An upregulated gene is a gene whose expression increases under a particular condition compared with the control. In your analysis, this is represented by a positive log₂ fold change.
7. What is a downregulated gene?
 - A downregulated gene is a gene whose expression decreases under a particular condition compared with the control. It is represented by a negative log₂ fold change.
8. Why is normalization needed when comparing RNA-seq samples?
 - Normalization accounts for differences in the total number of reads and other differences between samples. This makes it more appropriate to compare gene expression levels across samples.
9. Why is adjusted p-value commonly used instead of only the raw p-value?
 - RNA-seq tests thousands of genes at the same time, so some genes could appear significant by chance. The adjusted p-value helps account for these multiple comparisons; your analysis used adjusted p-value < 0.05 as the significance threshold.
10. Why might your results differ from those reported by the original authors?
 - The results may differ because of differences in software, analysis parameters, sample number, reference genome, or annotation files. Even small changes in the analysis workflow can affect the final list of differentially expressed genes.
11. How can changes in gene expression help an organism respond to environmental stress or infection?

- Changes in gene expression allow cells to produce more or less of certain molecules needed to respond to a stress or infection. For example, genes involved in immune responses can change their expression to help the organism respond to invading pathogens.

12. Does increased RNA abundance always mean that the amount of functional protein also increases? Explain.

- No. Increased RNA does not always result in increased functional protein because RNA can be degraded, translated at different rates, or affected by post-transcriptional regulation. Protein activity can also be changed after translation.

Conclusion:

The analysis successfully showed that infection can influence gene expression in the host, with clear differences observed between the control and infected samples. Some of the genes identified in our analysis were also found in the original study, indicating that our results were able to observe similar biological responses to *Staphylococcus aureus* infection. Although differences in the analysis tools, reference files, and settings may have affected the specific results, the main pattern of gene-expression changes was still observed. This activity showed us that RNA-seq is useful for studying how organisms respond to infection and that careful selection of analysis methods is important when reproducing results from published studies.